Reece M. Dobro

rmdobro@ncsu.edu | Waxhaw, NC | (980)-347-6003 | reecedobro.onrender.com

Education

NORTH CAROLINA STATE UNIVERSITY **EXP. MAY 2028**

Bachelor of Science in Electrical Engineering
Cumulative GPA: 4.00 | Dean's List: Fall 2024 - Present

Projects

AUTOMATED DOWNTIME REPORT SYSTEM | REFRESCO BEVERAGES **JUNE 2026 - AUG 2026**

- Independently conceptualized and engineered a full PLC-to-SCADA framework across 4 PLCs for a $17M, 1,500 CPM line
- Developed root-cause algorithms to identify electrical and instrumentation faults; reduced undeclared stops by 92%
- Field-tested PLC-to-SCADA fault detection, validating instrumentation and machine safety signals against machine behavior
- Built dynamic VBA reporting macros and authored extensive system documentation to ensure operational adoption

MAXIMUM POWER POINT TRACKER | NC STATE SOLARPACK **DEC 2025 - APRIL 2026**

- Led a 7-member team in redesigning the MPPT PCB, increasing power conversion efficiency by 12%
- Updated power electronics components and electrical schematic to support a 35% increase in solar array capacity
- Conducted end-to-end validation of MPPT hardware, analyzing full system performance under the FREEDM Systems Center
- Integrated CAN bus protocols to enable real-time telemetry exchange between custom decentralized vehicle boards

Professional Experience

ELECTRICAL ENGINEERING INTERN | REFRESCO BEVERAGES **MAY 2026 - AUG 2026**

- Audited one-line schematics across a production line, mapping 1,000+ connections and verifying electrical specifications
- Configured and tested PLC logic routines through Studio 5000 to improve system reliability and reduce unplanned downtime
- Analyzed electrical system faults to identify instrumentation failures, eliminating repeat downtime and saving $24,000
- Engineered a SCADA-linked high-speed imaging system to find root downtime causes in 150°F inner-machine areas.

PROCESS ENGINEERING INTERN | WIELAND COPPER PRODUCTS **MAY 2025 - AUG 2025**

- Engineered a complete automated cleaning cart, programming PLC logic for precise water, air, and IPA sequencing
- Designed cart power distribution and backup power for motors and solenoids while reducing projected build costs by 20%
- Reverse engineered an industrial Autosaw Head and fabricated replacement components, reducing costs by 77%
- Analyzed Mazak Megastir process data with Python to identify failure trends and diagnose root causes

Leadership

HIGH VOLTAGE LEAD | NC STATE SOLARPACK **AUG 2024 - PRESENT**

- Integrated vehicle electrical systems, connecting power, telemetry, and control subsystems to support reliable operation.
- Led 12 engineers in developing EV HV systems, integrating BMS, contactors, protection circuits, and safety controls
- Mitigated ground-loop degradation from capacitive coupling by auditing J1772 timing specs and redesigning isolation paths
- Implemented CAN-based power monitoring to track voltage, current, and power across vehicle systems

LOGISTICS ORGANIZER & SYSTEMS ENGINEER | HACKATHON NC STATE **AUG 2025 - PRESENT**

- Developed a 400+ participant tracking system to quantify key user metrics and identify improvement opportunities
- Selected RFID-based identification and engineered a portable tracking system for 60+ hours of operation
- Evaluated user feedback to improve system reliability, earning praise from major sponsors for its innovative design

WEBMASTER | NC STATE ENGINEER'S COUNCIL **DEC 2024 - PRESENT**

- Independently created a new website from the ground up in 2 weeks, translating leadership goals into a cohesive platform
- Incorporated feedback into technical updates, coordinating priorities and delivering solutions under tight deadlines

Skillset

Power & Electrical: HV Systems, Electrical Schematics, Power Monitoring, Troubleshooting, Instrumentation, PCB Design

Controls & Automation: PLCs, Ladder Logic, Structured Text, SCADA, Studio 5000, RSLinx Classic, Machine Safety

Software & Programming: Altium, KiCad, LTspice, MATLAB, AutoCAD, Inventor, WaveForms, Python, C/C++

Certifications: Autodesk AutoCAD, Autodesk Inventor, CSWA, FCC Amateur Radio, Microsoft Excel, Python MTA